

P0942R0: Introducing a `<smart_ptr>` header

Introduction

The current `<memory>` header has a mix of tools for two widely different audiences:

- smart pointers, recommended from day 1 to avoid explicit new/delete
- allocators and uninitialized storage, for experts

It was a mistake to mix tools for these two audiences in a single header. I therefore suggest to introduce a new header `<smart_ptr>` where all the smart pointers live. For backward compatibility, `<memory>` must include `<smart_ptr>`.

Discussion

This has been raised in passing several times, so here is the paper that actually does it.

Regarding feature testing, checking for the presence of the `<smart_ptr>` header suffices.

Wording

Add `<smart_ptr>` to table 16 in subclause 20.5.1.2 [headers].

Create a new first subclause [smart_ptr.syn] under 23.11 [smartptr]:

The header `<smart_ptr>` defines several smart pointer types and various function templates that operate on objects of these types.

Move all declarations for entities described in [smartptr] from [memory.syn] to [smart_ptr.syn]:

```
// 23.11.1, class template unique_ptr
template<class T> struct default_delete;
template<class T> struct default_delete<T[]>;
...
// 23.11.8, atomic smart pointers
template<class T> struct atomic<shared_ptr<T>>;
template<class T> struct atomic<weak_ptr<T>>;
```

Change 23.10.2 [memory.syn] paragraph 1 as follows:

The header `<memory>` defines several types and function templates that describe properties of pointers and pointer-like types, manage memory for containers and other template types, destroy objects, and construct multiple objects in uninitialized memory buffers (23.10.3-23.10.11). ~~The header also defines the templates `unique_ptr`, `shared_ptr`, `weak_ptr`, and various function templates that operate on objects of these types (23.11).~~

Add a new subclause in Annex D [depr]:

D.x Deprecated location of smart pointers [depr.memory.smartptr]

The header <memory> has the following addition:

```
#include <smart_ptr>
```